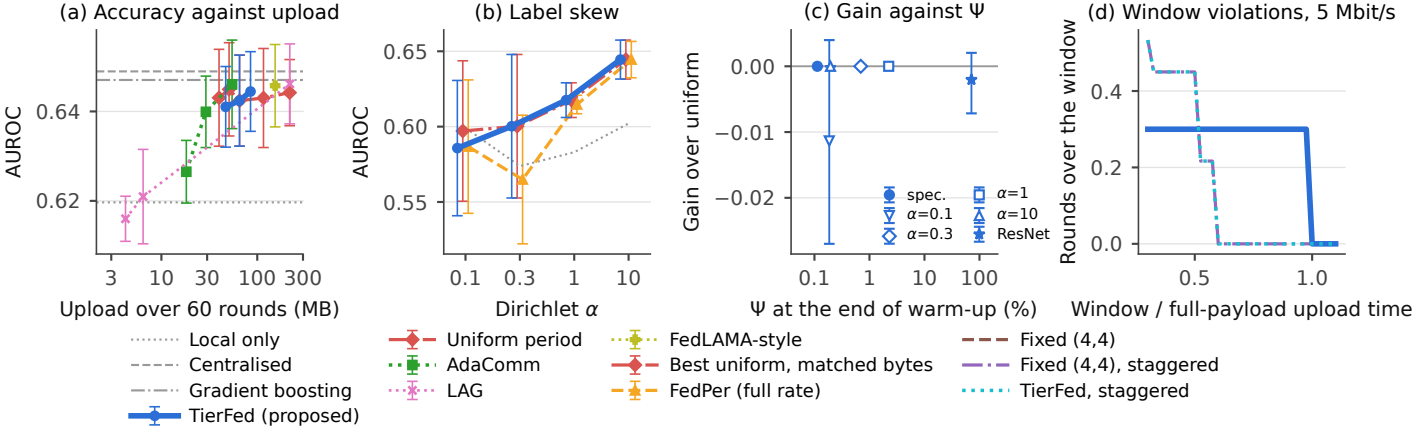
(a) Accuracy against upload
AUROC
0.64
0.62
3
10
30
100
300
Upload over 60 rounds (MB)
(b) Label skew
AUROC
0.65
0.60
0.55
0.1
0.3
1
10
Dirichlet α
(c) Gain against Ψ
Gain over uniform
0.00
−0.01
−0.02
0.1
1
10
100
Ψ at the end of warm-up (%)
spec.
α=0.1
α=0.3
α=1
α=10
ResNet
(d) Window violations, 5 Mbit/s
Rounds over the window
0.4
0.2
0.0
0.5
1.0
Window / full-payload upload time
Local only
Centralised
Gradient boosting
TierFed (proposed)
Uniform period
AdaComm
LAG
FedLAMA-style
Best uniform, matched bytes
FedPer (full rate)
Fixed (4,4)
Fixed (4,4), staggered
TierFed, staggered